Public Safety Surveillance System using Deep Learning

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Abstract— **In today's day and age, public safety is a matter of huge concern. Every day, a lot of people are subjected to different kinds of abuse, from verbal to physical. In most cases, by the time the crime is reported and the authorities arrive at the scene of the crime, the assaulter has already fled the scene. The aforementioned is only the case of reported crimes. A large number of criminal acts go unreported and the victim stays hushed out of fear. This brings the urgent need to automate violence and non-violence detection in public areas.**

The Public Safety Surveillance System is the perfect solution to this problem. This paper discusses the working of the system based on leveraging deep learning technology. It also explains the use of MobileNetv2 which is a lightweight state-of-the-art deep learning architecture. Apart from the functioning of the project, it also presents evaluation metrics of the system. Towards the end, there is a mention of the applications and the future scope of the project.

Keywords- Image augmentation, MobileNetv2, Deep Learning, Model Evaluation.

I. INTRODUCTION

In today's society, ensuring safety and preventing violence is paramount. The Public Safety Surveillance System is a component of security and safety measures. It is an automated platform designed specifically to identify instances of violence in video content. Its core functionality is based on the utilization of a deep learning architecture called MobileNetV2.
MobileNetV2's training on diverse datasets enhances our violence detection system's versatility across computer vision tasks. Leveraging this model enables accurate differentiation between violent and non-violent instances, facilitating prompt intervention and appropriate response. However, real-life scenarios may introduce challenges like image clarity variations and external conditions, potentially affecting the algorithm's precision.

A Comprehensive Violence Detection System

In light of these challenges, our research project proposes the development of a comprehensive Violence Detection System. This system leverages the pre-trained knowledge within MobileNetV2 and introduces image enhancement algorithms, all implemented within the versatile programming language, Python. By fusing the capabilities of deep learning with the strengths of MobileNetV2, our aim is to create a solution that is both reliable and efficient in the task of flagging violent behaviour.

Objective and Motivation

The project aims to accurately classify videos as violent or non-violent by extracting frames, expanding the dataset, and employing MobileNetV2 with machine learning. Evaluation metrics like precision, recall, and F1 score will assess model accuracy, driven by the goal of enhancing public safety and preventing violent incidents.
The motivation driving this venture is attached to the need to enhance public safety. This project aims to develop a surveillance system that accurately detects violence, enabling prompt intervention by law enforcement or security personnel to mitigate harm and prevent violent incidents.

II. RELATED WORK

The research community has shown significant interest in the automated detection of violence and non-violence in different settings, mostly propelled by its critical effect on cultural wellbeing and security. Consequently, numerous investigations in existing literature are cantered around the automated detection of violent and non-violent activities in various contexts.

In [1] , authors review state-of-the-art techniques for violence detection in surveillance videos. The comprehensive analysis categorizes these techniques into three distinct categories: Traditional machine learning, including methods such as KNN and Adaboost, SVM which are a set of supervised learning methods effective for classification, regression, and outliers detection, and Deep Learning. SVMs are particularly highlighted for their advantages, demonstrating effectiveness in high-dimensional spaces and robust performance even when the number of dimensions exceeds the number of samples. The research delves into various aspects of these methods, encompassing feature extraction techniques, object detection approaches and the utilization of video features. Furthermore, the paper provides an insightful architecture diagram to enhance understanding.

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Consider [2], In the context of this paper, computer vision strategies are employed for perceiving human activities, with an emphasis on violence detection in videos. It addresses the constraints of prior strategies, highlighting issues like optical flow-based problems and sensitivity to changes in illumination, presents a more robust approach using feature descriptors around interest focuses, and plans to assess the presentation of action recognition techniques in different video settings. The system includes making a new dataset from NHL hockey games, utilizing spatio-worldly descriptors like STIP and MoSIFT, applying a Sack-of-Words structure, and utilizing SVMs to group recordings as “violence” or “non-violence”.

The study [3], reviews the significance of CCTV camera video streams in the context of big data applications within surveillance. It features the normal utilization of deep learning methods for undertakings such as object and action recognition, crowd analysis, and violence detection. In any case, the paper also brings up that ongoing challenge and existing techniques have their limits. The study introduced in the paper offers an exhaustive outline of related research and outlines future directions pointed toward overcoming these challenges.

The paper [4] compares late advances in object identification and face recognition for video surveillance frameworks with regards to precision and speed. Quicker R-CNN with Inception ResNet V2 accomplishes high accuracy in real-time rates. Single Shot Finder (SSD) with MobileNet is quick and precise for most applications. FaceNet with Multi-task Cascaded Convolutional Networks (MTCNN) is quicker and more precise than DeepFace and DeepID2+. A start to finish video surveillance framework is proposed, and potential future enhancements are examined, including video object identification and video salient object detection approaches.

[5] This study examines the rapid evolution of machine learning, specifically deep learning since 2006. Deep learning emulates brain neurons' computational processes, advancing artificial intelligence significantly. It enables automated image and text recognition, facilitating real-time human-machine interactions and has been integrated into security systems for high-definition video data processing. The authors also give an outline of the preparation in fostering a campus video security monitoring system. The research highlights deep learning’s crucial role in enhancing campus security monitoring systems and outlines central standards and ideas for such systems, ultimately establishing the overall architecture for implementation. Integrating deep learning algorithms enhances campus security by improving accuracy, emergency response, operational efficiency, standards, and overall safety and stability.

The authors of [6] have significantly contributed to Video Salient Object Detection (VSOD) by introducing the Densely Annotated VSOD (DAVSOD) dataset, involving 226 recordings with 23,938 casings and top notch explanations. The research addresses the challenge of saliency shift in videos, evaluating 17 VSOD algorithms on their DAVSOD dataset, the largest benchmark in the field, using three performance metrics. The study introduces a novel convLSTM network model, addressing saliency shift challenges, and highlights the significant DAVSOD dataset, offering key insights for future VSOD research and development.

The research in [7] presents a critical headway in melanoma detection, detection, using deep learning and machine learning with the MobileNetV2 network. This lightweight design enables potential deployment on cell phones. Experimental results on various melanoma datasets uncover a noteworthy exactness of up to 85%, outperforming other networks. The head model's design, featuring a global average pooling layer and two fully-connected layers, strikes a balance between high accuracy and network efficiency. This examination can possibly change early melanoma recognition, a pivotal move toward diminishing death rates. It offers a useful answer for accurate diagnosis, particularly while considering the effectiveness expected for accurate diagnosis. The proposed model for low-complexity violence detection in [8], uses CNN-VGG16. It offers a reliable approach for automated violence detection. The model makes use of a pre-trained model, CNN-VGG16. It leverages its architecture by using its pre-existing knowledge in detecting shapes and edges. This in turn helps reduce the number of parameters that need to be trained and improves the efficiency of the network. The model was evaluated on two datasets, each containing violent videos and achieved a high accuracy rate of almost 99% for the dataset that had a Canny filter on. The Canny filter is an edge detection algorithm that is generally used to highlight the most significant features in an image which was discovered to improve performance. The proposed model can potentially be used in real-time video streams and incorporate additional features such as audio and optical flow to help prevent and detect violence. However, it requires more research and a more diverse dataset.

[9] delves into license plate color recognition using a combination of transfer learning and the lightweight MobileNetV2 architecture. By integrating Transfer Learning, the study tackles the challenge of balancing dataset size and accuracy, effectively reducing reliance on large datasets. MobileNetV2's efficiency and configurability make it ideal for deployment on mobile and embedded devices, offering low latency and efficient operation. Despite training on a modest dataset, the model achieved an impressive 99% accuracy, indicating its potential for real-world applications in parking management, traffic control, and urban planning. Field tests further validate the model's performance, highlighting opportunities for future improvements and optimizations.

[10] explores the application of Deep Learning, specifically MobileNetV1 and MobileNetV2, in improving breast cancer diagnosis by analysing histopathological images. MobileNetV1 exhibited better validation accuracy but also overfitting compared to MobileNetV2. The study highlights the importance of early and accurate breast cancer detection, emphasizing the potential of deep learning techniques in enhancing diagnostic efficiency. Despite limitations in previous studies due to small datasets, the authors propose an automated diagnosis system to address this challenge. Additionally, the research underscores the significance of accurately detecting invasive ductal carcinoma (IDC) regions within whole-mount slides to streamline diagnosis processes and reduce costs. MobileNetV1's depth-wise separable convolution and MobileNetV2's residual structure contribute to their effectiveness in various applications, with MobileNetV1 demonstrating superior detection accuracy in this project.

III. PREREQUISITES

i. Deep learning: It is a subset of machine learning, revolves around multi-layered neural networks, spanning from feedforward for structured data to CNNs for images and RNNs for sequential data.

ii. MobileNetV2: It plays a vital role in this project in terms of image classification and feature extraction. It requires a deep understanding of its architectural components, including Depth Wise separable convolutions and inverted residuals, which contribute to model efficiency and reduced computational costs. MobileNetV2, a lightweight model with just 32 layers, offers significant computational power for CNN processes, making it ideal for real-time image analysis and processing.

iii. Python: Python is a high-level programming language known for its simplicity, versatility, and readability, making it widely used in various fields such as web development, data analysis, artificial intelligence, and more. It serves as the cornerstone of this project, bridging theory with practice. Mainly used libraries in python for this project are TensorFlow, Keras, CV2, Sklearn, Matplotlib, ImageIO and Numpy.

iv. TensorFlow: It is an open-source deep learning framework developed by Google, widely used for building and training machine learning models. It provides a flexible ecosystem for implementing various neural network architectures and conducting large-scale computations efficiently. TensorFlow plays a central role in the project, requiring proficiency in core concepts like tensors and computation graphs, as well as designing neural network architectures and training models. Understanding these nuances is crucial for effectively harnessing TensorFlow's deep learning capabilities.

v. Keras: Keras is a user-friendly deep learning framework. It acts as a valuable ally in this project and offers a streamlined path from theory to practice. Due to its pythonic syntax and modular design, it simplifies the development of neural networks reducing the gap between idea and implementation. With its pre-trained models, Keras accelerates research and application development, enabling users to leverage existing knowledge and expertise. This framework caters to both industrial and academic researchers. It offers a smooth and productive research experience.

vi. GPU Support: They are specialized processors designed for accelerating image rendering and performing parallel computations across multiple data sets simultaneously. Originally developed for gaming, GPUs have evolved into highly parallel processors with massive computational power, finding applications in fields like scientific computing, machine learning, deep learning and data analysis.

vii. CV2 : It serves as a powerful toolbox enabling computers to perceive and analyze images and videos, similar to how people manipulate visuals with tools. Widely utilized in AI and robotics, it's the go-to for tasks like facial recognition and various image and video manipulations. Its compatibility with popular languages like Python serves as a handy assistant for all image and video-related tasks in the computer world.

IV. PROPOSED METHODOLOGY

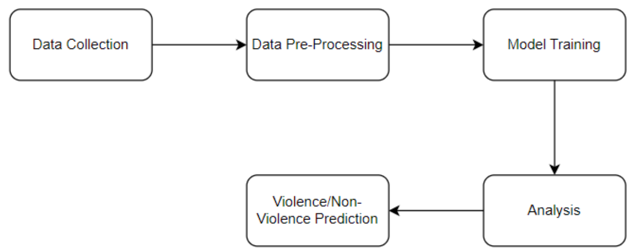

FIG 1: Block diagram for implementation

i. Data Collection: In the initial phase of the project, data is gathered from various sources, such as surveillance cameras, social media feeds, and other relevant sensors. These sources provide raw data that is important for subsequent analysis. The collected data comes in various formats and structures, making it crucial to ensure a seamless and comprehensive process.

ii. Data Pre-processing: The collected data needs to be pre-processed and transformed into a suitable format for analysis. In this project, we introduce image augmentation as a pre-processing technique. Image augmentation entails applying a series of transformations to images in order to diversify the training data, This in turn helps in improving the model's ability to generalize from the dataset and make accurate predictions.

iii. Model Training: This stage of the project involves training a machine learning model, particularly neural networks in context of deep learning, to learn from the given dataset and be able to make accurate predictions. Model training begins with pre-processing the data, compiling it with appropriate settings and adjusting the model's parameters to improve its effectiveness and produce a model that can generalize well and make accurate predictions.

iv. Analysis: The model is then tested using several metrics and the results of these metrics are plotted visually.

v. Violence/Non-Violence Predictions: The last step of the project is feeding a video to the model and testing whether or not the model aptly predicts the nature of the behaviour being displayed.

V. IMPLEMENTATION

This section of the research paper discusses in detail the functioning of the project. It will cover the following topics: About dataset, Model building and Prediction.

A. About the dataset:

The dataset consists of 1000 real time violence and non-violence videos each. The non-violent dataset contains videos ranging from day-to-day activities caught on surveillance footage to friendly gestures practiced by humans on a daily basis. The authors of the data set enacted a couple of videos to make sure that the dataset is diverse in nature. The dataset also has a set of videos taken from different movie scenes. The violence dataset consists of a cusp of real-time violence caught on camera and violence scenes from various movies. This helps obtain a more diverse dataset for better accuracy. For the sake of the experiment, 300 violent and 300 non-violent videos were taken into consideration.

B. Model Building:

The model leverages deep learning techniques and state of the art architectures to achieve high accuracy in violence detection. The model building process has three crucial parts, namely:

I. Pre-Processing
II. Model Training
III. Model Evaluation.

This section mainly focuses on the aforementioned topics.

I. Pre Processing : Pre-Processing mainly refers to the process of transforming and preparing the data before feeding it into a deep learning model. It helps improve the quality and reliability of the input data. This essentially improves the performance and effectiveness of the deep learning model. It involves the following key steps:

i. Frame Extraction: The video frames are extracted from the dataset. The OpenCV library helps read a video file and extract each frame.

ii. Data Augmentation: To improve the versatility and the diversity of the dataset, data augmentation techniques are applied to the dataset. It helps improve the robustness and generalization of the dataset. The augmentation techniques include frame flipping, zooming, adjusting brightness, and rotation. Upon augmenting the dataset, the video capture is released using vidcap.release() and the list of frames is returned as the output of the function. The frames are resized to a standardized dimension. (128x128 pixels).

iii. Data Splitting: The data is then split into training and testing keeping the class distribution constant. Reshaping and normalization is done to prepare the input feature such that its format is suitable for training the neural network.

II. Model Training : Model training is the process of training neural networks to learn from the training dataset and make accurate predictions. It usually involves defining the model's architecture and its settings, iteratively tuning the model's parameters through training and with the help of back propagation and optimization algorithms. The aim is to produce a model that can generalize well and accurately predict outcomes.

i. Model Architecture: In this project, a pre-trained model MobileNetV2 is used as the backbone. It is fine tuned for violence detection by adding a custom head with a single output unit using a sigmoid activation function. The binary classification setup is chosen to distinguish between the violence and non-violence.

ii. Transfer Learning: Transfer learning is implemented to leverage the knowledge gained by the model on a large scale dataset. The base model's layers are kept frozen to retain the features of the pre-trained model and only the custom head is trained to adapt the model to the task required to perform.

iii. Hyperparameter tuning: Several hyperparameter tuning techniques are used, such as, learning rate, batch size, and early stopping criteria. These are set to optimize the training process. The learning rate is implemented in such a way that it gradually increases and then decreases the learning rate during training.

iv. Callback functions: Multiple callback functions are implemented during training, including early stopping, model checkpointing and learning rate reduction on plateau, ensuring stable and efficient training of the model.

III. Model Evaluation : Model evaluation serves as a critical gauge of a model's performance and its applicability in the real world. It is done to assess how well a model generalizes unseen data, identifies potential shortcomings and validates the effectiveness of the chosen algorithm and hyperparameters. The model evaluation consists of the following phases:

i. Predictions: The model is used to predict the violence labels for the test datasets. The threshold for the predictions are set at 0.5 to classify each video as violent or non-violent.

ii. Metrics and Analysis: Several classification metrics are computed such as, accuracy, precision, recall, F1-score, and the confusion matrix. These metrics are discussed briefly in the following section. This step is vital because it provides insight into the model's stability and performance and also its ability to correctly classify the videos as violent or non-violent.

iii. Visualization: The easiest way to interpret data is through visualization. The results are visualized using libraries like matplotlib and seaborn. The confusion matrix and classification report are presented for a detailed analysis of the performance.

iv. Model Saving: Upon evaluation, the model trained model is saved for future deployment and development.

C. Prediction:

Prediction is a fundamental concept in data analysis which involves forecasting or estimating an outcome or value based on existing data and patterns. It majorly relies on the historical data and the identification of relationships between variables to perform informed projections about future events. It helps make more sound predictions and data-driven decisions.

This section of the project classifies the video as either violent or nonviolent. It loads the pre-trained models for violence detection, which extracts frames of the input data and performs predictions on each frame.

VI. Experimental Results

Performance of the developed model is rigorously evaluated using the presented evaluation metrics for this purpose. The metrics allow for a fair assessment of the study results and allow for their useful comparison against the baseline models. The evaluation metrics applied in the project includes - Confusion matrix, F1 score, Recall and Precision.

i. Confusion matrix: The confusion matrix is a tool for evaluating the performance of a machine learning model especially when dealing with classification tasks. It categorizes the models predictions into four categories based on the actual, namely - True Positives (TP), True Negatives (TN), False Positives (FP) and False Negatives (FN).

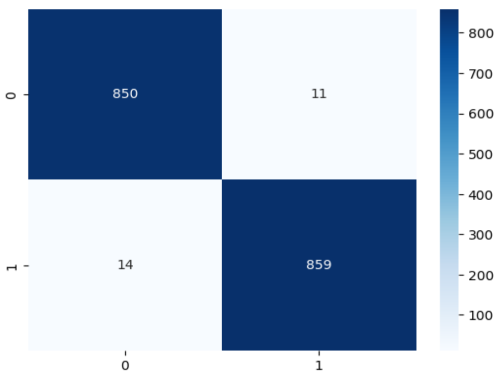

FIG 2. Confusion matrix

i. F1 Score: 2 * (Precision * Recall) / (Precision + Recall)
ii. Recall = TP / (TP + FN)
iii. Precision = TP / (TP + FP)

TABLE I. Values of the metrics

F1 Score	0.985
Recall	0.9839
Precision	0.987

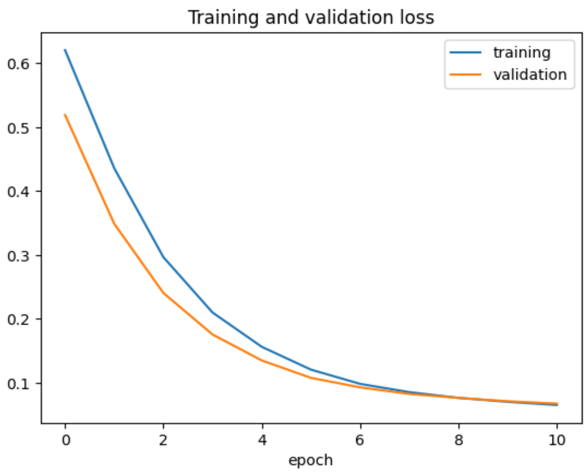

FIG 3. Training and validation loss

TABLE II. Values of training and validation loss

Epochs	Training loss	Validation loss
2	0.35	0.25
4	0.19	0.15
6	0.13	0.11
8	0.09	0.085
10	0.085	.08

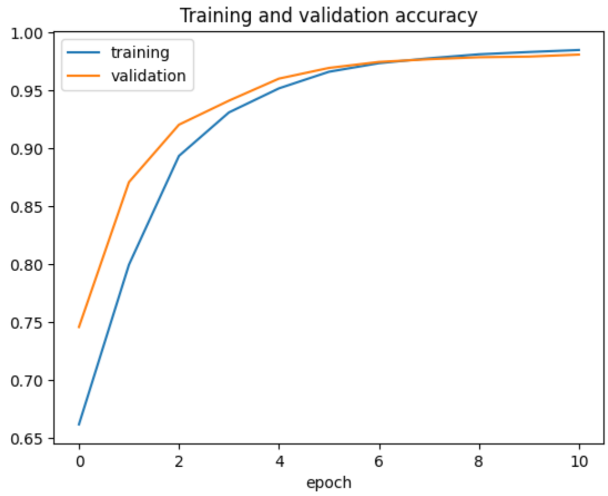

FIG 4. Training and validation accuracy

TABLE III. Values of training and validation accuracy

Epochs	Training accuracy	Validation accuracy
2	0.88	0.90
4	0.95	0.95
6	0.958	0.956
8	0.96	0.956
10	0.963	.96

FIG 5. Non-violence detection

In FIG 5, the algorithm predicts a non-violent classification based on the absence of indicators of bodily harm or aggressive facial expressions. Through analysis of relative body movements, the algorithm discerns the lack of violent cues, leading to its classification as non-violent. The binary prediction as “Violence: False” indicating non-violence is depicted on the top left corner of the frame.

FIG 6. Violence detected

In FIG 6, the algorithm predicts a violent scenario due to the presence of objects exhibiting aggressive bodily movements and clear indications of violent facial expressions. The binary prediction “Violence: True” is displayed in the top left corner of the frame, affirming the presence of violence as identified by the algorithm.

The algorithm employs analysis of object motions, activities, gestures, including facial expressions such as anger, sadness, and happiness, as well as motion sensing of relative body movements to classify incidents as violent or non-violent.

VII. Applications

Real-time violence detection has found valuable applications in enhancing the safety and security of the public. Some of its currently used real-time applications are:

i. Enhancing Public Security: Crowd Management and Urban Surveillance.
ii. Educational Environments: School Security and Bullying Prevention.
iii. Transportation and Infrastructure Safety: Public Transport Security and Critical Infrastructure Protection.
iv. Content Moderation and Online Safety: Content Filtering and Cyberbullying Prevention.
v. Emergency Response Assistance: First Responder Support and Disaster Management.

VIII. Future Scope

The project has several potential avenues for future expansion and development including but not limited to:

i. Real-time monitoring.
ii. Multi-modal Analysis.
iii. Privacy Preservation.
iv. Community Feedback Integration.
v. Collaboration with law enforcement.
vi. Anomaly Detection.

In the bigger picture, the future scope of the project involves taking the technology to higher levels, diversifying its use, and implementing it in practical scenarios to enhance security and safety.

IX. Conclusion

The development of a violence detection system, harnessing the MobileNetV2 pre-trained model and sophisticated image enhancement algorithms within a Python framework, represents a significant advancement in the fields of public safety and surveillance. After rigorous training, the system achieved an impressive 96% accuracy, demonstrating its proficiency in accurately distinguishing between violent and non-violent content.

Notably, the model's computational efficiency is a standout feature. Its streamlined design and low resource requirements make it an excellent choice for continuous applications and devices with limited resources. This ensures seamless integration into various platforms, such as surveillance systems and mobile applications, without overburdening hardware resources.

Beyond its technical achievements, this project serves a dual purpose. Firstly, it enhances public safety by providing a robust tool for violence detection, aiding law enforcement and digital content providers in mitigating violent incidents. Secondly, it contributes to understanding digital violence challenges, offering insights for policy-making and social initiatives, ultimately fostering a more peaceful society. This proactive approach to comprehending and addressing violence is a substantial contribution to the broader goal of fostering a more peaceful and violence-free society.

In essence, this project stands as a versatile and impactful solution that not only elevates public safety but also enhances our understanding of the multifaceted issues surrounding violence in the digital age. It exemplifies the potential of technology to have a positive societal impact and represents a symbol of progress in the fields of public safety and surveillance.

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